



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -3

CO5_AT3

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Course code : CSA0708

Course name: Computer Networks

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MOCK INTERVIEW

1. Analyze how network congestion affects packet delay and packet loss.

- Congestion occurs when traffic exceeds network capacity.
- Packets wait longer in router queues, increasing delay.
- Queue overflow causes packet loss.
- Retransmissions further increase congestion.
- Overall network performance decreases.

2. Compare the impact of star and mesh topologies on network reliability and fault tolerance.

Star Topology

1. Easy to install.
2. Failure of one cable affects only one device.
3. Central switch failure stops the network.
4. Low cost.
5. Easy troubleshooting.

Mesh Topology

1. Multiple paths between devices.
2. Very high reliability.
3. Excellent fault tolerance.
4. Expensive to implement.
5. Suitable for critical networks.

3. Why does increasing the number of routers increase end-to-end delay?

1. Each router processes packets.
2. Store-and-forward delay occurs.
3. Queueing delay increases.
4. More propagation distance.
5. Total delay becomes higher.

4. Analyze how QoS improves voice and video traffic.

1. Prioritizes real-time traffic.

2. Reduces delay.
3. Minimizes jitter.
4. Lowers packet loss.
5. Improves call and video quality.

5. Evaluate fiber optic vs copper cables.

1. Fiber provides higher bandwidth.
2. Supports longer distances.
3. Immune to electromagnetic interference.
4. More secure.
5. Higher installation cost than copper.

6. Analyze VLANs for performance and security.

1. Reduces broadcast traffic.
2. Improves bandwidth utilization.
3. Separates departments logically.
4. Enhances security.
5. Simplifies network management.

7. Compare TCP and UDP.

TCP

1. Reliable.
2. Error checking.
3. Flow control.
4. Slower.
5. Used for web and email.

UDP

1. Faster.
2. No retransmission.
3. Low overhead.
4. Less reliable.
5. Used for VoIP and gaming.

8. Importance of load balancing.

1. Distributes network traffic.
2. Prevents server overload.
3. Improves availability.
4. Increases performance.
5. Enhances fault tolerance.

9. Effect of frequent routing updates.

1. Consumes bandwidth.
2. Increases CPU usage.
3. Keeps routing information updated.
4. Improves route accuracy.
5. Excessive updates reduce efficiency.

10. Security measures vs performance.

1. Firewalls inspect packets.
2. Encryption consumes CPU.
3. Security adds processing delay.
4. Protects sensitive data.
5. Balance between security and speed is needed.

11. High delay—identify and fix.

1. Use Ping and Traceroute.
2. Check router CPU usage.
3. Monitor bandwidth utilization.
4. Remove congestion.
5. Upgrade network if required.

12. Low throughput despite high bandwidth.

1. Packet loss.
2. High latency.
3. Network congestion.
4. TCP retransmissions.

13. Reduce packet loss.

1. Increase bandwidth.
2. Apply QoS.
3. Upgrade routers/switches.
4. Reduce congestion.
5. Replace faulty cables.

14. VoIP call breaking frequently.

1. Check packet loss.
2. Measure jitter.
3. Verify delay.
4. Configure QoS.
5. Check internet connection.

15. Improve high jitter.

1. Enable QoS.
2. Reduce congestion.
3. Increase bandwidth.
4. Use wired connection.
5. Optimize routing.

16. Simulate congestion in Packet Tracer.

1. Build a network.
2. Generate heavy traffic.
3. Limit bandwidth.
4. Observe packet delay.
5. Analyze Simulation Mode.

17. Packets delayed in a router.

1. Check CPU utilization.
2. Examine queue size.
3. Verify routing table.
4. Upgrade router if needed.

18. Optimize network for online gaming.

1. Reduce latency.
2. Minimize jitter.
3. Use wired Ethernet.
4. Enable QoS.
5. Close background downloads.

19. Design a small network and measure performance.

1. Connect PCs through a switch.
2. Add router and server.
3. Test using Ping.
4. Measure RTT and throughput.
5. Observe packet loss.

20. Diagnose high RTT.

1. Run Ping.
2. Use Traceroute.
3. Check congestion.
4. Inspect routing path.
5. Upgrade slow links.

21. Explain network performance to a non-technical person.

1. Like road traffic.
2. Data are vehicles.
3. Delay is traffic jam.
4. Speed is throughput.
5. Good roads mean better performance.

22. Throughput real-life example.

1. Water pipe analogy.
2. Pipe size is bandwidth.
3. Water flow is throughput.

4. Blockages reduce flow.
5. Wider pipe gives more throughput.

23. Packet loss affects video streaming.

1. Video freezes.
2. Buffering increases.
3. Poor video quality.
4. Audio-video mismatch.
5. Interrupted viewing experience.

24. Explain delay types.

1. Processing delay.
2. Queueing delay.
3. Transmission delay.
4. Propagation delay.
5. Total delay is the sum of all.

25. Describe Cisco Packet Tracer.

1. Cisco network simulation tool.
2. Designs virtual networks.
3. Configures routers and switches.
4. Tests networking concepts.
5. Used for CCNA learning.

26. Explain QoS simply and technically.

1. Gives priority to important traffic.
2. Reduces delay.
3. Controls bandwidth.
4. Uses traffic classification.
5. Improves voice and video performance.

27. Present simulation results to a client.

1. Show network topology.
2. Explain objectives.

3. Present graphs and metrics.
4. Compare before and after optimization.
5. Give recommendations.

28. Explain jitter with analogy.

1. Like buses arriving irregularly.
2. Some arrive early.
3. Some arrive late.
4. Timing becomes inconsistent.
5. Voice quality suffers.

29. Testing network performance in a lab.

1. Created network in Packet Tracer.
2. Used Ping.
3. Measured delay.
4. Checked packet loss.
5. Compared performance after optimization.

30. Importance of performance metrics.

1. Measures network health.
2. Detects problems.
3. Improves user experience.
4. Supports optimization.
5. Ensures reliability.

31. Introduce yourself.

1. State your name.
2. Mention education.
3. Interested in networking.
4. Skilled in Packet Tracer.
5. Ready to learn and contribute.

32. Interest in networking and simulation.

1. Networking connects the world.
2. Simulation avoids hardware costs.
3. Improves practical skills.
4. Helps solve real problems.
5. Supports career growth.

33. Confidence in analyzing network performance.

1. Understand networking concepts.
2. Use Packet Tracer.
3. Analyze metrics.
4. Solve troubleshooting problems.
5. Continuous learning.

34. Strong networking concept.

1. TCP/IP model.
2. Layer functions.
3. Packet transmission.
4. Routing process.
5. Real-world applications.

35. Handling unknown questions.

1. Stay calm.
2. Admit honestly.
3. Explain related concepts.
4. Say you will learn.
5. Maintain confidence.

36. Solved a technical problem.

1. Identified the issue.
2. Used troubleshooting tools.
3. Found the root cause.
4. Implemented the solution.

37. Why should we select you?

1. Strong networking basics.
2. Good problem-solving skills.
3. Quick learner.
4. Hands-on Packet Tracer experience.
5. Positive attitude and teamwork.

38. Challenges while learning Packet Tracer.

1. Initial configuration errors.
2. IP addressing mistakes.
3. Routing configuration issues.
4. Learned through practice.
5. Improved troubleshooting skills.

39. Improve technical skills regularly.

1. Practice daily.
2. Watch tutorials.
3. Read networking books.
4. Build lab simulations.
5. Earn certifications.

40. Explain a complex concept simply.

1. Start with a simple example.
2. Avoid technical jargon.
3. Use real-life analogies.
4. Explain step by step.
5. Confirm listener understanding.